Unregrettable Distributed Power Control in Over-the-Air Computation Systems

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***Abstract*—6G wireless networks are shifting from data-centric communication to integrated computation. Over-the-Air Computation (AirComp) supports the ultra-low latency data aggregation in massive Internet of Things (IoT) ecosystems. However, AirComp technology is hindered by the communication bottlenecks of centralized scheduling and the challenge of achieving coherent signal superposition under dynamic channel conditions. In this paper, we address this issue by introducing a novel "unregrettable" distributed power control framework based on non-cooperative game theory and regret learning. We formulate the power allocation problem as a non-cooperative game where the devices autonomously optimize their transmission power levels in order to minimize the aggregation error at the receiver. The convergence to a stable Correlated Equilibrium is proven without requiring global channel state information. Our approach introduces a unique utility function with a channel-aware projection mechanism and an adaptive inertia parameter that allow the system to robustly handle high dynamic range channels and accelerate convergence. Extensive simulations demonstrate that our distributed framework outperforms traditional centralized approaches in terms of execution time, improved energy efficiency, and scalability, with comparable aggregation error.**

***Index Terms*—Over-the-Air Computation, Distributed Power Control, Regret Learning, Correlated Equilibrium.**

I. Introduction

Modern wireless 6G networks and IoT have moved from the purely data-centric transmission to an integrated environment of communications and computation. Following this paradigm, traditional Orthogonal Multiple Access (OMA) schemes become obsolete given that the wireless resources scale in a linear manner with respect to the number of devices, thus creating a significant communication bottleneck in ultra-dense wireless networks, where low latency data aggregation is critical [1]. Recently, Over-the-Air Computation (AirComp) emerged as a transformative technology to address this scalability issue by exploiting the superposition property of the wireless multiplex channel and allowing multiple devices to transmit simultaneously on the same frequency [2]. In this paper, the operation of the AirComp technology is optimized by introducing a regret learning-based non-cooperative game-theoretic approach enabling the devices to optimize their transmission power levels in order to ultimately optimize the reception of the superimposed signal at the receiver by minimizing the aggregation error without the overhead of complex central scheduling.

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A. Related Work

Recent research works have focused on the distributed optimization and machine learning workflows within the AirComp technology. Specifically, AirComp has been shown to reduce the bandwidth consumption for general distributed convex optimization problems when it is compared to traditional digital communication strategies [3]. Multi-cell AirComp systems have been studied in [4], dealing with the spectrum sharing, while considering the interference between the multiple cells and aiming at maximizing both the throughput and the accuracy of the aggregated computation.

A major challenge in implementing the AirComp technology is the coherent superposition of the multiple signals over the air in order for the target function to be accurately received. This goal can be achieved by sophisticated *power control mechanisms* in order to invert the channel fading effects and optimally align the signals' magnitude and phase at the receiver [5]. Several convex optimization techniques have focused on the minimization of the Mean Squared Error (MSE) problem in multi-cell setups by jointly optimizing the beamforming at the access point and the power allocation at the devices' level [6]. These techniques are characterized by high computational and signaling overhead in order to gather the channel state information. Recently, Graph Neural Networks have been used to support the distributed power allocation, where the wireless network is formulated as a graph and messages are exchanged between the nodes/devices in order to acquire the channel state information [7]. Also, Deep Reinforcement Learning has been used to support the simultaneous wireless information and power transfer in AirComp systems by optimizing the resource allocation and maximizing the long-term energy efficiency [8].

AirComp technology can support a wide range of computation-based applications in wireless networks [9], while the Federated Learning (FL) is the most compelling use case given its inherent characteristic of relying on the aggregation of high-dimensional parameter vectors [10]. Several recent techniques have used the inherent channel noise within the AirComp-based communication as a way to enhance privacy by masking the individual gradient updates [11]. Also, efforts have been devoted on studying the AirComp-based communications for FL applications with time varying wireless channels and limited channel state information [12]. However, the existing approaches are mainly based on centralized optimization.

B. Contributions & Outline

Though substantial research efforts have been devoted in the power control problem to realize the AirComp technology, the existing literature mainly relies on centralized optimization frameworks or complex deep learning models that introduce high signaling overheads and extreme computational demands on resource-constrained devices. Also, the existing approaches mainly address the power control problem as a static one, while also requiring perfect global channel state information. Additionally, the initial distributed schemes lack convergence guarantees or they converge to suboptimal local equilibria, thus, they introduce significant aggregation errors when the channel state conditions fluctuate in dynamic wireless communication environments.

Aiming at addressing those research gaps, we introduce a novel regret learning-based non-cooperative game-theoretic power control framework characterized by the distributed decision-making among the devices and introducing a novel paradigm shift from the rigid centralized scheduling. The main contributions of this research work are summarized as follows.

1) A novel power control problem in AirComp-based systems is formulated as a non-cooperative game and a uniquely designed utility function is introduced to incentivize the devices to balance their individual transmission power and ultimately minimize the aggregation error.
2) A low-complexity distributed learning algorithm is designed in order to enable the devices to iteratively adapt their transmission power strategies accounting for the regret that they experience from past actions. The proposed algorithm eliminates the need for a centralized computation approach and determines the optimal transmission power levels with a lower execution time.
3) The overall framework guarantees the convergence of all the devices to a Correlated Equilibrium and provides a robust and strictly superior solution compared to the centralized optimization approach with respect to improving energy efficiency and convergence time, while achieving comparable aggregration error.
4) Additionally, the proposed solution guarantees the long-term system stability and efficiency given that the devices participate in an unregrettable resource management approach, which ensures that they will not prefer a different power strategy compared to the power vector of the Correlated Equilibrium.
5) Detailed simulations demonstrate the superiority of the proposed approach compared to state-of-the-art centralized and distributed benchmarks in terms of achieving lower transmission power and improved convergence speed and comparable aggregation error, particularly in large scale systems.

Section II presents the AirComp model. The distributed power control based on regret learning, the existence of a Correlated Equilibrium, and the convergence of the algorithm to it are presented in Section III, while Section IV analyzes the numerical results. Section V concludes the paper.

II. AirComp Model

An AirComp-based system is considered consisting of a set $\mathcal{N} = \{1 \dots, n, \dots N\}$ of edge devices that act as transmitters, and a single edge server that receives the devices' data. Based on the AirComp technology, the receiver can detect several computed functions, e.g., arithmetic mean, max, min, etc.. In this paper, we focus on the arithmetic mean without loss of generality. Each device n generates a normalized data symbol $s_n \in \mathbb{R}$ that is modeled as a random variable with zero mean and unit variance, i.e., $\mathbb{E}[s_n] = 0$ and $\mathbb{E}[s_n^2] = 1$. The target aggregated signal f is defined as follows:

$$f = \frac{1}{N} \sum_{n \in \mathcal{N}} s_n. \tag{1}$$

The devices' signals need to be optimally synchronized in terms of their phase to minimize the aggregation error at the receiver. However, in realistic wireless communication environments, the transmitted signals suffer from channel fading and noise. Thus, the devices need to optimally pre-compensate their transmissions in order to achieve the minimization of the aggregation error. The superimposed signal is derived as:

$$y = \sum_{n \in \mathcal{N}} |h_n| \sqrt{P_n} s_n + W, \tag{2}$$

where $|h_n|$ denotes the channel gain magnitude between the device n and the receiver and $W \sim \mathcal{N}(0, \sigma^2)$ is the Additive White Gaussian Noise (AWGN).

The devices' superimposed signal is received by the edge server, which then applies a denoising scaling factor η to recover the target function. The estimated recovered signal $\hat{f}$ is calculated as follows:

$$\hat{f} = \frac{y}{N\sqrt{\eta}}. \tag{3}$$

Aiming at quantifying the performance of the overall AirComp-based system, the Mean Squared Error (MSE) metric is adopted. The MSE quantifies the distortion between the estimated signal $\hat{f}$ and the target signal f. Considering that in a realistic wireless communication environment, the data symbols $\{s_n\}$ are independent and uncorrelated with the noise W, the MSE is derived as a function of the power allocation vector $\mathbf{P} = \{P_n\}_{n \in \mathcal{N}}$ and the denoising scaling factor η as:

$$\begin{aligned}
\text{MSE}(\mathbf{P}, \eta) &= \mathbb{E}\left[\left(\hat{f} - f\right)^2\right] \\
&= \frac{1}{N^2} \mathbb{E}\left[\left(\frac{y}{\sqrt{\eta}} - \sum_{n \in \mathcal{N}} s_n\right)^2\right] \\
&= \frac{1}{N^2} \mathbb{E}\left[\left(\sum_{n \in \mathcal{N}} \left(\frac{|h_n|\sqrt{P_n}}{\sqrt{\eta}} - 1\right) s_n + \frac{W}{\sqrt{\eta}}\right)^2\right] \\
&= \frac{1}{N^2} \sum_{n \in \mathcal{N}} \left(\frac{|h_n|\sqrt{P_n}}{\sqrt{\eta}} - 1\right)^2 + \frac{\sigma^2}{N^2 \eta}.
\end{aligned} \tag{4}$$

III. Unregrettable Distributed Power Control in AirComp Systems

Considering that the receiver has fixed hardware characteristics, the denoising scaling factor is considered fixed. Therefore, the only control parameter is the power allocation of the devices. Aiming at minimizing the aggregation error (Eq. 4), we formulate the interactions among the devices as a non-cooperative game $\mathcal{G} = (\mathcal{N}, \mathcal{P}, \{U_n\}_{n\in\mathcal{N}})$, where $\mathcal{N}$ is the set of players, i.e., devices, $\mathcal{P} = \prod_{n\in\mathcal{N}} \mathcal{P}_n$ is the overall strategy set, where the strategy space of player n, $\mathcal{P}_n$, represents the set of discrete power levels $\mathcal{P}_n = \left\{0, \frac{P_n^{\max}}{L}, \frac{2P_n^{\max}}{L}, \ldots, P_n^{\max}\right\}$, with $P_n^{\max}$ [W] denoting the maximum transmission power of each device n, and L is the quantization level.

The uniquely designed utility function U_n for player n, given the strategies of other players $\mathbf{P}_{-n}$, is defined as:

$$U_n(P_n, \mathbf{P}_{-n}) = \frac{g(|h_n|)\sqrt{P_n}}{\sqrt{\eta}} \sum_{n'\in\mathcal{N}, n'\neq n} \left(\frac{g(|h_{n'}|)\sqrt{P_{n'}}}{\sqrt{\eta}}\right) - \frac{g(|h_n|)^2 P_n}{\eta} \sum_{n'\in\mathcal{N}, n'\neq n} \left(\frac{g(|h_{n'}|)^2 P_{n'}}{\eta}\right) \quad (5)$$

where

$$g(|h_n|) = \frac{1}{|\log|h_n||^{\alpha}} \quad (6)$$

with $\alpha \in \mathbb{R}^+$ denoting a control parameter. The physical meaning of the utility function U_n captures the incentivization of the devices to optimally align their transmitted signal magnitudes by selecting power allocations that maximize their utility. Also, the projection function $g(|h_n|)$ accounts for the high dynamic range of the wireless channels by appropriately compressing them for the devices' consideration during the selection of the optimal transmission power levels. As observed by the utility function U_n, the decision of one device regarding its transmission power P_n is coupled with the decisions of the rest of the devices $\mathbf{P}_{-n}$. Moreover, perfect channel state information is not needed, as each device only requires the overall signal stemming from the rest of the devices within the network, which can be easily broadcasted by the edge server without requiring personalized information for each device.

A. Correlated Equilibrium

The goal of each device is to maximize its utility function by distributively deciding its optimal transmission power level. To achieve this goal, we introduce a distributed learning approach, where the non-cooperative game is played iteratively over discrete time slots $t = 1, 2, \ldots$ and the corresponding action profile of the devices is denoted as $\mathbf{P}^t = (P_1^t, \ldots, P_N^t)$. The goal of the devices is to converge to an equilibrium that is stable and that computationally achievable.

Existing approaches in the literature adopt the Pure Nash Equilibrium (PNE), which however is characterized by significant limitations, i.e., it may not exist, and it is often computationally intractable to convert to it. Therefore, in this paper, the concept of Correlated Equilibrium (CE) is adopted, which is a generalization of the PNE. The physical meaning of the CE is that it acts as a signaling mechanism, where the devices receive recommendations regarding their strategies according to a joint distribution. Each device has no incentive to deviate from the assigned signal, assuming that the other devices follow the recommendations.

Considering that $\Delta(\mathcal{P})$ denotes the set of probability distributions over the joint strategy space $\mathcal{P}$, then the CE of the game $\mathcal{G}$ is formally defined as the probability distribution $\boldsymbol{\chi} \in \Delta(\mathcal{P})$, if for every player $n \in \mathcal{N}$ and every pair of distinct strategies $a_j, a_k \in \mathcal{P}_n$, the following condition holds [13]:

$$\sum_{\mathbf{P}\in\mathcal{P}:P_n=a_j} \boldsymbol{\chi}(\mathbf{P})[U_n(a_k, \mathbf{P}_{-n}) - U_n(\mathbf{P})] \leq 0. \quad (7)$$

The physical meaning of Eq. 7 is that if a device n receives a recommendation to play a_j, then they cannot improve their utility by deviating to a different strategy a_k considering that the recommendations are drawn from the probability distribution $\boldsymbol{\chi}$ and the rest of the devices do not deviate from the recommendations. The set of CE is a convex set that contains the set of Nash Equilibria, i.e., $NE \subseteq CE$ [13].

B. Regret Learning-based Distributed Power Control

Aiming at determining the CE in a distributed manner, we introduce a regret learning-based approach. Each device calculates its regret for not having played other strategies in the past to ultimately make the optimal decision regarding its transmission power level that maximizes its utility (Eq. 5) and reduces the aggregation error (Eq. 4). Let $H_t = (\mathbf{P}^1, \ldots, \mathbf{P}^t)$ denote the history of the game up to the time slot t.

For any two strategies $a_j, a_k \in \mathcal{P}_n$, let $W_n^{\tau}(a_j, a_k)$ be the counterfactual utility that player n would have received at time τ if they had played a_k instead of a_j, assuming that all other players' actions $\mathbf{P}_{-n}^{\tau}$ remained fixed:

$$W_n^{\tau}(a_j, a_k) := \begin{cases} U_n(a_k, \mathbf{P}_{-n}^{\tau}), & \text{if } P_n^{\tau} = a_j, \\ U_n(\mathbf{P}^{\tau}), & \text{otherwise.} \end{cases} \quad (8)$$

Then, the average regret of the device n at time t for not replacing strategy a_j with a_k is defined as the difference between the average counterfactual utility and the actual utility:

$$D_n^t(a_j, a_k) := \frac{1}{t}\sum_{\tau=1}^{t} (W_n^{\tau}(a_j, a_k) - U_n(\mathbf{P}^{\tau})) = \frac{1}{t}\sum_{\tau\leq t:P_n^{\tau}=a_j} \left[U_n(a_k, \mathbf{P}_{-n}^{\tau}) - U_n(\mathbf{P}^{\tau})\right]. \quad (9)$$

Considering that each device is interested only in potential improvements of its utility, we derive the positive regret:

$$R_n^t(a_j, a_k) := \max\{D_n^t(a_j, a_k), 0\}. \quad (10)$$

Each device n considers its experienced regret values and determines its strategy probability distribution $\boldsymbol{\psi}_n^{t+1} \in \Delta(\mathcal{P}_n)$

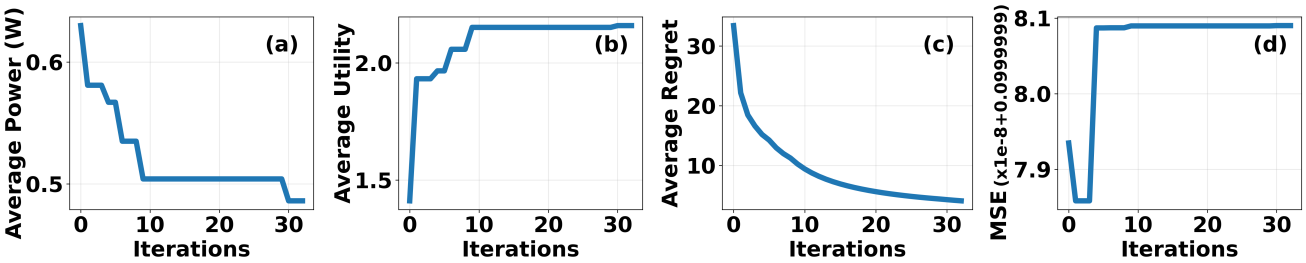

Fig. 1: Convergence of devices' average (a) transmission power, (b) utility, (c) regret, and (d) MSE at the CE.

at time slot $t+1$. By denoting with $a_j = P_n^t$ the action chosen in the previous round, then, the probability of switching to a different strategy a_k is proportional to the regret of not having played this action and is given as follows:

$$\psi_n^{t+1}(a_k) := \frac{1}{\mu} R_n^t(a_j, a_k), \quad \forall a_k \neq a_j. \tag{11}$$

The probability of retaining the same strategy a_j is the residual probability that is derived as follows:

$$\psi_n^{t+1}(a_j) := 1 - \sum_{a_k \neq a_j} \psi_n^{t+1}(a_k). \tag{12}$$

where $\mu > 0$ is a sufficiently large inertia parameter chosen to ensure $\psi_n^{t+1}(a_j) > 0$.

The fundamental convergence property of the regret learning process is established by Hart and Mas-Colell [13].

Theorem 1. *If every player $n \in \mathcal{N}$ follows the regret learning process, the empirical distribution of the joint actions, denoted by $\boldsymbol{x}^t$, converges almost surely as $t \to \infty$ to the set of Correlated Equilibria of the game $\mathcal{G}$. The empirical distribution $\boldsymbol{x}^t \in \Delta(\mathcal{P})$ is defined as the relative frequency of the joint strategies played over the horizon t:*

$$\boldsymbol{x}^t(\mathbf{P}) := \frac{1}{t} \sum_{\tau=1}^{t} \mathbb{I}(\mathbf{P}^\tau = \mathbf{P}), \tag{2.3}$$

where $\mathbb{I}(\cdot)$ is the indicator function.

The distributed regret learning for AirComp power control algorithm is presented in Algorithm 1.

Algorithm 1 Distributed Regret Learning for AirComp Power Control

Input: N, η, $\{h_n, P_n^{\max}\}_{n\in\mathcal{N}}$, σ^2, convergence threshold ϵ, inertia parameter μ.
Output: Empirical Equilibrium Distribution $\boldsymbol{x}^*$.
Initialize: $t = 1$, *Converged* = false.
Initialize probability vectors $\boldsymbol{\psi}_n^1 = \{\frac{1}{L}, \ldots, \frac{1}{L}\} \, \forall\, n \in \mathcal{N}$.
Initialize cumulative regret matrices $\mathbf{D}_n = \mathbf{0}_{L\times L} \, \forall\, n$.
Initialize empirical distribution map $\boldsymbol{x}^0$.
while not *Converged* **do**
  Step 1: Action Selection
  for each device $n \in \mathcal{N}$ **do**
    Select action P_n^t according to distribution $\boldsymbol{\psi}_n^t$.
  end for
  Construct joint action profile $\mathbf{P}^t = (P_1^t, \ldots, P_N^t)$.
  Step 2: Update Empirical Distribution
  Update count: $\boldsymbol{x}^t[\mathbf{P}^t] \leftarrow \boldsymbol{x}^{t-1}[\mathbf{P}^t] + 1$.
  Step 3: Update Regrets
  for each device $n \in \mathcal{N}$ **do**
    Let $a_j = P_n^t$ (action played).
    for each alternative strategy $a_k \in \mathcal{P}_n$ **do**
      Calculate instantaneous regret:
      $\Delta U = U_n(a_k, \mathbf{P}_{-n}^t) - U_n(\mathbf{P}^t)$.
      Update average regret (recursive Eq. 9):
      $D_n^t[j,k] = \frac{t-1}{t} D_n^{t-1}[j,k] + \frac{1}{t}\Delta U$.
      Compute positive regret: $R_n^t[j,k] = D_n^t[j,k]^+$.
    end for
  end for
  Step 4: Update Strategies
  for each device $n \in \mathcal{N}$ **do**
    Let $a_j = P_n^t$.
    for each $a_k \in \mathcal{P}_n, a_k \neq a_j$ **do**
      $\psi_n^{t+1}(a_k) = \frac{1}{\mu} R_n^t[j,k]$.
    end for
    $\psi_n^{t+1}(a_j) = 1 - \sum_{k\neq j} \psi_n^{t+1}(a_k)$.
  end for
  Step 5: Check Convergence
  if $||\frac{\boldsymbol{x}^t}{t} - \frac{\boldsymbol{x}^{t-1}}{t-1}||_\infty < \epsilon$ **then**
    Converged = true.
  end if
  $t \leftarrow t + 1$.
end while

IV. Numerical Results

In this section, detailed simulations are presented for the proposed distributed regret learning-based non-cooperative game-theoretic power control framework. Section IV-A presents the pure operation of the proposed framework in term of energy efficiency and aggregation error. Section IV-B demonstrates a thorough scalability analysis, while Section IV-C analyzes the impact of the wireless communication environment on the proposed framework. Section IV-D signifies the impact of adaptive learning on the proposed framework, while Section IV-E provides a thorough comparative evaluation. The set of parameters adopted in our simulations are $P_n^{\max} = 1W, \forall n \in \mathcal{N}$, the nodes are distributed in a radius 100m, and the higher ID node associated with the receiver is characterized by the highest distance, $1,000$ Monte Carlo trials are carried out, the channel gain follows the Outdoor Urban Microcell (UMi, Street Canyon), and Indoor Hotspot (InH, Office) scenarios as defined in 3GPP TR 38.901 [14], $L = 100$, $n = 10$ or $n = 100$, $\sigma^2 = 10^{-12}$, $f_c = 2.405$ GHz, $\eta = 0.5$, $\alpha = 0.1$ $\epsilon = 0.005$ and $\mu = 3000$, unless otherwise explicitly stated. The evaluation was performed on a Precision 3680 Computer with Intel Core i7 14700 @2.10GHz processor, 64 GB available RAM.

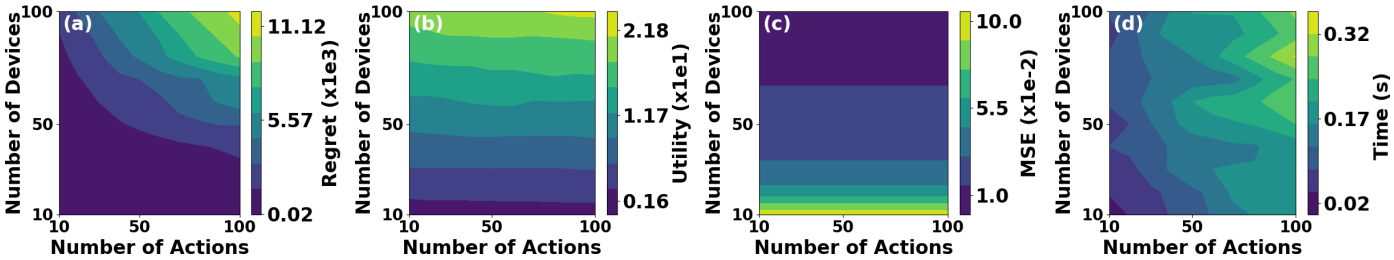

Fig. 2: Scalability analysis: Devices' (a) regret, (b) utility, (c) receiver's MSE and (d) system's execution time.

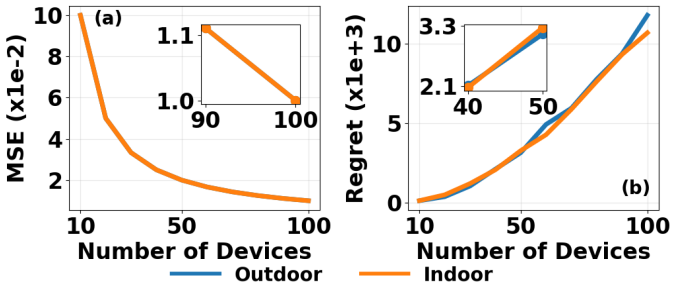

Fig. 3: Outdoor and Indoor (a) MSE, and (b) regret

A. Distributed Power Control

In this section, the performance and operation of the proposed distributed regret learning-based non-cooperative game-theoretic power control framework is presented for an AirComp system consisting of 10 devices and $\mu = 500$. Fig. 1a–1d illustrate the devices' average power, utility, and regret, as well as the MSE as a function of the iterations, respectively, in order for Algorithm 1 to converge to the CE. The results reveal that the devices manage to self-organize while minimizing their individual regret (Fig. 1c) and maximizing their utility (Fig. 1b) by appropriately selecting their optimal transmission power (Fig. 1a) through the regret learning process. Therefore, the devices effectively align their signal magnitudes in order to minimize the aggregation error at the receiver (Fig. 1d).

B. Scalability Analysis

In this section, a scalability analysis of the proposed framework is presented by varying the number of devices and the number of discrete power actions. Fig. 2a – 2d present the heat maps of the devices' total regret, utility, the receiver's MSE, and the system's execution time as a function of the number of devices and number of actions, i.e., discrete power levels, respectively. The results show that the MSE decreases as the number of devices and actions increase (Fig. 2c). This reduction in MSE stems from the coherent superposition of the signals when more devices participate in the transmission, as the correlated noise and the channel distortions are averaged out. Simultaneously, the increasing number of actions provides the devices with a finer power control granularity that minimizes the quantization error between the ideal and the selected power levels. Furthermore, the results reveal the persistent low execution time of the proposed framework even for large scale systems with fine power control, which proves that the AirComp technology can overcome the signaling overheads that limit the centralized scheduling in ultra-dense 6G networks (Fig. 2d). It is observed that the increasing number of actions contribute to a small increase in the devices' utility (Fig. 2b), as they can more accurately select their optimal transmission power. The increase in the devices' utility results in also a small increase of the regret (Fig. 2a), as the latter one is defined as the difference in utility considering all the different strategies.

C. Impact of Communication Environment

In this section, we analyze the system's performance under two distinct 3GPP channel models, i.e., Outdoor Urban Microcell and Indoor Hotspot [14]. Fig. 3a–3b present the MSE and the devices' regret as a function of the number of the devices for the two scenarios, respectively. In both scenarios, the proposed regret learning approach achieves very similar behavior (both in terms of the receiver's MSE and the devices' regret) due to its adaptability in the channel gain characteristics. The proposed framework enables the devices to choose an optimal transmission power level in order to maintain coherent signal superposition and low aggregation error without requiring perfect global channel state information, which is often difficult to obtain in a realistic wireless environment.

D. Adaptive Regret Learning Analysis

In this section, the regret learning approach is analyzed for a constant and adaptive inertia parameter μ considering an AirComp system with 10 devices. In the adaptive scenario, inertia parameter μ is dynamically updated based on the inverse of the total regret of each device at each time slot. Fig. 4a–4e present the histograms of the constant and adaptive μ regret learning algorithms' iterations, receiver's MSE, devices' average utility, power, and total regret, respectively. The results reveal that the adaptive mechanism optimizes the exploration-exploitation trade-off. In the early stages, a lower inertia encourages the devices to more thoroughly search the strategy space for better power levels, and prevents them from getting trapped in suboptimal local equilibria. As the system approaches the Correlated Equilibrium and the regret diminishes, the inertia increases, and the network stabilizes. This dynamic self-tuning leads to a more responsive system that can quickly (Fig. 4a) lock into an optimal configuration that achieves low MSE values (Fig. 4b) and enhances the overall efficiency of the AirComp system in terms of energy efficiency (Fig. 4d), and the devices' experienced utility (Fig. 4c) and regret (Fig. 4e).

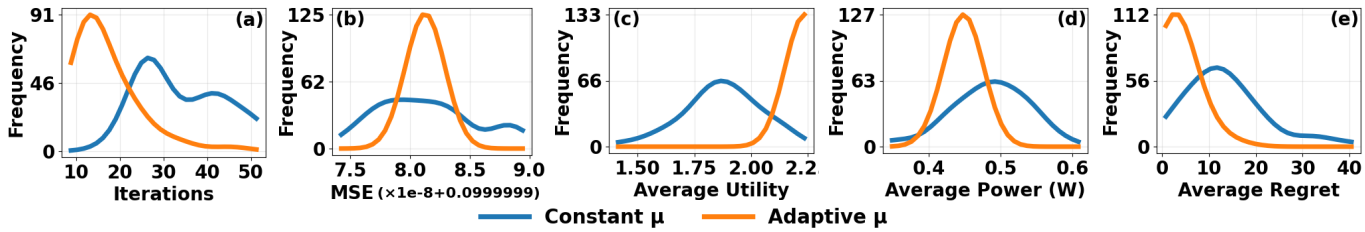

Fig. 4: Regret learning with constant and adaptive inertia parameter μ: (a) iterations, (b) MSE, (c) average utility, (d) average transmission power, and (e) average regret at the CE.

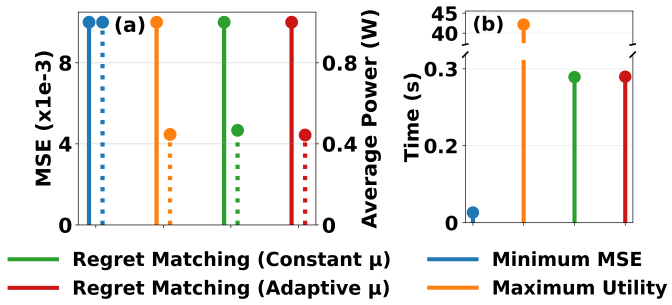

Fig. 5: Comparative analysis (a) MSE and average power (solid line: MSE, dotted line: Power), and (b) time.

E. Comparative Evaluation

A thorough comparative evaluation of the proposed framework is performed compared to the following scenarios: (i) Min MSE, (ii) Max Utility, where a centralized problem is solved to minimize the MSE and maximize the summation of the devices' utilities respectively, and regret learning-based power control with (iii) constant and (iv) adaptive inertia parameter μ, considering a large-scale system of 100 devices. The results clearly demonstrate the superiority of the proposed adaptive distributed regret learning-based non-cooperative game-theoretic power control framework in terms of achieving similar MSE and improved energy efficiency (Fig. 5a), as well as execution time to converge to the Correlated Equilibrium (Fig. 5b). It is noted that even though the centralized methods theoretically offer the perfect power allocation, they are practically limited by the need to collect global channel information, which induces high latency. The proposed distributed method achieves an "unregrettable" performance in terms of enabling the devices to maximize their own utility and have no incentive to change their equilibrium strategies, thus, achieving an optimal solution with substantially decreased execution time. The proposed approach can effectively solve the scalability and latency issues of next generation networks.

V. Conclusions

In this paper, a pioneering distributed power control framework for AirComp systems is introduced based on non-cooperative game theory and regret learning. A novel utility function is proposed to capture the devices' dynamics within the wireless communications environment and a non-cooperative game is formulated among them in order to determine their optimal transmission power levels that ultimately decrease the aggregation error at the receiver. A distributed regret learning mechanism is designed to converge to the non-cooperative game's Correlated Equilibrium. Detailed numerical results demonstrate the superiority of the proposed approach compared to existing centralized power control solutions. Part of our current work focuses on the testing of the proposed framework at the PROTON Lab testbed available at Arizona State University consisting of 2,000 wireless connected nodes.

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